Push 2 / Nuendo Bridge

Release Notes

Version 1.0.8 — October 2026

New Features

- **Chord display**: the name of the chord you hold on the pads is shown on screen — in the Mix / Pan footer, or as a small badge in the top-right corner of every other page. Triads, sixths, sevenths, ninths, 11ths, 13ths, sus and power chords, shell voicings without the fifth, and inversions (“C/E”). Sharps or flats follow the active scale; a Setup preference (MIDI Ctrl tab) decides when the scale doesn't (Piano layout, Chromatic, C major…).
- **Velocity range and sensitivity**: the Vel Curve tab gains **Sens** (Encoder 6: how hard you must hit to reach full velocity), **Min** and **Max** (Encoders 7 / 8: the velocity sent by the softest and the hardest hit). The curve previews follow the settings live.
- **Count-in**: **Shift + Metronome** toggles Nuendo's count-in (precount). While it is on, the Metronome button blinks — slowly when the metronome is off, fast when it is on.
- **Nuendo 15 Group Folders**: Mute, Solo, Monitor and Record now work on Group Folder tracks from the Push.
- **Calmer list scrolling**: the encoders now step more slowly through lists and numbers so you can land on an item — MIDI CC numbers, XY parameters, Plugin Browser collections / folders / plugin list, and keyswitch notes.

Bug Fixes

- **“NuendoBridge Out” kept returning to “In ‘All MIDI Inputs’”** (macOS): every bridge launch looked like a brand-new port to Nuendo, so its MIDI Port Setup choices were lost and the bridge's control messages could reach MIDI tracks and change their parameters. The bridge's ports now keep the same identity from one launch to the next, so the setting sticks.
- **Recording with pre-roll**: started from the Push, recording now waits for the end of the pre-roll instead of punching in almost immediately.
- **Velocity curves**: the curves now span the pad pressure range actually used when playing — previously even a hard hit only reached the low part of the curve, so high velocities were out of reach. The saved curve is also re-applied when the bridge starts (the Push forgets it when powered off).
- **About page**: the script version no longer shows “not connected” when the bridge is started after Nuendo.

Technical

- macOS: the four virtual MIDI ports get a stable CoreMIDI unique ID (kMIDIPropertyUniqueID derived from the port name). Windows uses loopMIDI ports, which were already persistent.
- Record is fired as the `Transport / Record` key command instead of writing the host record value (which forced an immediate punch-in).

- Group Folders appear in the mixer as *GroupChannel*: Mute / Solo are written through DirectAccess (tags 1027 / 1028); Monitor / Record belong to the folder track and use the `Edit / Monitor` and `Edit / Record Enable` key commands on the selected folder (press event on CC 95 / 96, wire ch 8; acknowledgement CC 68 = 4 / 5).
- Expand-command table extended (39 Monitor, 40 Record Enable, 41 Precount On) — bridge and script tables kept in sync.
- The script re-sends its version every ~2 s; the bridge logs it only when it changes and clears it on disconnect.
- Velocity table: the Push indexes its 128-step table by pad force (0–4095 g); curves are now laid out from 160 g to the force selected by Sens (1700–400 g, default 1300 g). New settings keys: velocity_min, velocity_max, velocity_sens, chord_flats_default.
- Shared encoder slow-down helper (one step per 4 encoder ticks, 2 for the long plugin list).

Upgrade Note

- **macOS**: after installing 1.0.8, open Studio → Studio Setup → MIDI Port Setup **once** and uncheck “In ‘All MIDI Inputs’” for **NuendoBridge Out** (it appears as a new port the first time). From then on the choice is kept.
- As always, install the new **Ableton_Push2.js** together with the new bridge.

Known Limitations

- **Count-in LED**: Nuendo does not report the count-in state to MIDI Remote scripts, so the bridge only tracks its own Shift+Metronome presses. If you toggle the count-in in Nuendo, the LED will be out of step — toggle it once more in Nuendo to realign.
- **Group Folder Monitor / Record** select the folder in Nuendo (the key commands act on the selected track). The Push LEDs show what was toggled from the Push, not changes made in Nuendo.
- **Last track bank may show overlapping tracks** (unchanged since 1.0.7): when the track count isn't a multiple of 8, the final bank slides back to show the last 8 visible tracks.

Version 1.0.7 — July 2026

New Features

- **Key Commands**: the eight note-value buttons trigger any Nuendo Key Command, across 8 pages of 8 slots (Page Left/Right to switch). Per-slot colours, an on-screen legend on long-press, and a web editor (open the Plugin Mapper page, then the ■ *Key Commands* link, or go to /keycmd) with a searchable Nuendo function picker and one-click *Generate & install*.
- **Expression pedal**: passive expression pedals now work as continuous controllers on either jack, with a one-time on-screen calibration. Targets: MIDI CC, Volume, a Quick Control, a Send, or **AI Knob** (whatever the mouse hovers). The touchstrip LED bar follows the value.
- **Long-press expand menus**: hold New, Duplicate, Mute, Play, Record or Fixed Length to open an on-screen menu of related Nuendo commands on the upper/lower row. The menu stays until you switch display mode.
- **Plugin Browser overhaul**: browse by **folder** (your Plugin Manager collection folders), a vertical list showing each plugin's **name, type and manufacturer**, an option to group the Default collection by **Type** or **Vendor** (Setup → Browser), an instant on-disk cache, and it remembers your last-used collection.
- **Backup & Restore**: save and restore all bridge settings (Setup, pedals, Key Commands, plugin mappings, last collection) as a single zip — from the menu bar / tray or the Plugin Mapper web page.
- **Lower idle CPU**: the display now idles its refresh rate down when the screen is static, cutting the bridge's CPU use at rest by roughly two-thirds.
- **macOS**: *Start at Login* fixed; the app now requests Documents-folder access so *Generate & install* can update the installed script.

Technical

- Key Commands: generated `makeCommandBinding` per slot (note-value CCs 36-43, wire ch 12 / 0xBC); web editor endpoints under /api/keycommands + Nuendo Key Commands XML import; script generator writes the installed Ableton_Push2.js in place.
- Expand-menu commands on wire ch 13 (0xBD); command tables kept in sync between the bridge and the JS.
- Expression pedal via Push 2 display SysEx (0x30 contact config, 0x31 curve limits, 0x32 curve entries, 0x13 ADC sampling); calibration persisted per pedal.
- Plugin folder path (mCollectionPath) added to the plugin-list SysEx (0x2C); on-disk cache at ~/.push2bridge/plugin_list_cache.json; streaming batch raised to 40 entries/tick.
- Bank position is dead-reckoned with end-of-project detection; cells that never received a name render blank instead of a “Track NN” placeholder.
- Plugin parameter names are sanitised to ASCII (e.g. Unicode unit glyphs) before the 7-bit SysEx transport, and the Plugin Mapper name matcher normalises '.', '_' and spaces so decimal frequency bands match (e.g. Maag EQ4 “2.5 kHz”).
- Display loop always emits a frame (the Push blanks its screen after ~2s without one) but renders at an adaptive rate; macOS Info.plist gains NSDocumentsFolderUsageDescription; the macOS build zips with symlink preservation.

Known Limitations

- **Last track bank may show overlapping tracks.** Cubase's mixer bank zone always keeps 8 channels filled, so when the project's track count isn't a multiple of 8 the final bank slides back to show the last 8 visible tracks — overlapping the previous bank rather than showing a short, partial bank. The tracks shown are correct and fully controllable; only the grouping of the very last bank overlaps.

Version 1.0.6 — June 2026

New Features

- **Footswitch / pedals**: both Push 2 pedal jacks are now usable (jack 1 = CC 64, jack 2 = CC 69). Each pedal gets its own Setup page (Pedal 1 / Pedal 2) and can be assigned to *Sustain* (CC 64 passed through to the instrument), *Play*, *Stop*, *Play/Stop*, *Record*, *Rec/Stop* (start recording, press again to stop), or *Off*. An *Invert* option per pedal handles normally-closed switches. Defaults: Pedal 1 = Sustain, Pedal 2 = Play/Stop.
- **Persistent Setup settings**: aftertouch mode, velocity curve, fixed velocity, the Control-Room knob default and the footswitch configuration are now saved to `~/.push2bridge/settings.json` and restored on the next launch.
- **Navigation page** (Master button): overlays four directional D-pads on the 64 pads (corners) for Zoom, Scroll/cursor, Markers/locators and Nudge, each with hold-to-repeat. The D-pads stay active across the mix pages so you can navigate the timeline while you work; hold Master to flash an on-screen reminder. Note / Scale / Session restore the normal pads.
- **DirectAccess parameter capture** (Shift+Browse): capture a plugin's live parameters straight from Nuendo — for inserts **and** instruments — even for plugins the offline scanner can't read (iLok/PACE-protected, Waves shells, Nuendo's stock plug-ins). Capture feeds the Plugin Mapper directly, and an **Edit Map** button (Inserts parameter page) opens the Mapper on the selected plugin. The parameter cap was raised to 16384 to cover large instruments (2000+ parameters).
- **Track page redesign** (Shift+Mix): a single track-name header (like the Sends page); the upper row's buttons 3-8 toggle sends 1-6 on/off (buttons 1-2 inactive); the arrows navigate track by track.
- **Control Room Phones**: hold *Select* and turn the Master encoder to ride the headphone (Phones) bus instead of Main; the level shows in the Mix footer. The default target is configurable in Setup → CR Knob.
- **Clip button cycles Sends ↔ Pan** (the way Mix cycles Volume ↔ Channel Strip), and the top-left encoder now acts as Nuendo's **AI Knob** (controls whatever the mouse is hovering).
- **Plugin Mapper** improvements: correct default scan paths plus user-configurable folders (incl. Nuendo/Cubase stock VST3, version-independent), multi-plugin VST3 shell support, a scan progress bar with cancel / skip-current-plugin, per-plugin rescan, manufacturer/state/type filters, parameter search, drag-and-drop between slots, mapping export/import, and clearer error messages for Intel-only / PACE-protected plug-ins.

Bug Fixes

- **Bank desync in large projects fixed**: the bridge now receives live bank-position feedback from Nuendo (via DirectAccess on the bank zone) and self-corrects its bank offset — lost bank pulses, end-of-project clamping or tracks added/removed in Nuendo no longer leave bank 1 showing the wrong tracks. After a correction the visible bank's names and colours are re-sent so the display heals itself. Pressing ■ on bank 1 also re-pulses the bank zone to drag a drifted view back to the top.
- **MIDI CC collisions on channel 0 fixed**: the Navigation “Nudge Right” D-pad moved off CC 107 (which is Solo 8) to 108, and the DirectAccess Clear-All Monitor / Rec moved off CC 66/67 (which are send-enable 7/8) to 98/109 — so Shift+Mon / Shift+Rec no longer toggle a send.

- **Clear All Monitor / Record** now only act on channels that actually expose the Monitor / Record-Enable parameter — output busses, groups, VCA and the Control Room are skipped, so a clear no longer alters their volume/pan.
- **Channel Strip track navigation**: the arrows now move track by track in Channel Strip mode and re-explore the strip for the new channel.
- **Plugin parameter name matching rewritten** (multi-pass): exact matches can no longer be stolen by earlier fuzzy ones, single-word names match, unit suffixes are tolerated ('in_gain_db' ↔ 'In Gain') and truncated host names are recognized — fixes the cryptic “P12 ?” labels and dead encoders on console-style plug-ins (e.g. bx_Console SSL 4000 E). Parameter titles are also no longer cut at 20 characters.
- **Send next/prev no longer nudges pans**: the buttons shared CC 46/47 with the absolute pan bindings of bank tracks 7/8 — every press slammed a pan hard right. They moved to a dedicated channel, and repeated presses now always step (the second press used to be ignored).
- **Track-page send encoders are now relative**: the old absolute write could slam a send to min/max from a stale cached level (reported as “send jumped to +12”).
- **Volume / send-enable feedback moved to a dedicated MIDI channel**: on the old shared channel the volume feedback of tracks 3-8 was swallowed by other handlers, so faders moved in Nuendo snapped back when an encoder was next touched, and send-enable feedback could register as phantom peak-clips on tracks 1/2.
- **Muted tracks keep their colour** (dimmed) in the mixer header instead of turning a flat grey.
- **AI Knob** no longer collides with the send-enable button for track 5 (it moved off CC 64).

Maintenance

- Internal clean-up: removed dead code and unused helpers, and translated all remaining source comments to English.

Version 1.0.5 — May 2026

New Features

- **Keyswitch pad layouts**: the Layout button now cycles through four pad layouts — 64 notes, 56 notes + 8 keyswitches, 48 notes + 16 keyswitches, and Drum. Keyswitch pads sit on the bottom row(s), light orange, and trigger user-defined absolute notes (ideal for articulation keyswitches). Long-press Layout to configure: Enc 1 sets the starting note, Enc 2-8 override individual pads, and the lower row toggles Chromatic / Naturals layout, Latch (the keyswitch section is monophonic — only one active at a time), Reset, and Done. Note names follow Cubase convention (C-2 = MIDI 0). Configuration is in-memory (not persisted).
- **Two octatonic scales**: *Octatonic WH* and *Octatonic HW* added to the Scale menu.
- **XY pad** (Session button): the 64 pads become an XY morphing surface controlling two parameters of the selected track (Volume, Pan, Quick Controls 1-8) or raw MIDI CC. Relative, pressure-weighted input means no jump when you touch down, and two-finger input interpolates smoothly. Lower 1/2 pick the X/Y category (Track / CC), Enc 1/2 pick the X/Y parameter, Enc 4/5 set sensitivity and smoothing, Lower 5-8 toggle Mute/Solo/Monitor/Record for the selected track, and the arrows navigate tracks.
- **Pad note range in the Mix footer**: the Mix page footer now shows the MIDI note range currently mapped to the pads (keyswitch pads excluded in keyswitch layouts).
- **Per-track automation feedback**: each bank track's Read/Write automation mode is now reflected on screen, not just the selected track's.
- **Windows System Tray app**: on Windows the bridge now runs from the system tray (status, Open Plugin Mapper, Show Log, Quit) instead of a console window. Run with *--terminal* for the classic console version. The Plugin Mapper no longer opens a browser automatically.
- **Rescan moved to the Setup page** (lower-row button 8, available on every Setup page).
- **macOS app is now fully self-contained**: the libusb USB runtime is bundled inside the .app, so Homebrew and `brew install libusb` are no longer required — just copy the app and run it. (The pre-built .app is Apple Silicon only; Intel Macs run from source — see the User Guide.)

Bug Fixes

- **Control Room Main reset** (Shift+Touch on the rightmost encoder) now lands on an exact 0.00 dB instead of -0.01 dB.
- **Channel Strip module values** now appear immediately on entering a module (Gate, Comp, EQ, etc.) and stay visible, instead of only showing after you nudge a parameter.
- **Saturator and Limiter encoders** now respond to changes (the sub-page activation could fail silently for those two slots).
- **Inserts page**: an insert is no longer occasionally shown duplicated in another slot (display only).
- **CC Pick-up mode removed**: the MIDI CC page is Absolute only. Pick-up required parameter feedback that a normal setup does not provide, so it behaved identically to Absolute.

Version 1.0.4 — May 2026

New Features

- **Channel Strip mode** (Mix → Mix again): a dedicated Channel Strip view with an overview page (Gate, Comp, EQ, Tools, Sat, Limiter shown as Cubase-coloured banners) and six drill-down sub-pages — one per module.
- **Strip slot extended parameters**: parameters that are NOT in Cubase's bank zone are now reachable via DirectAccess on every strip variant — e.g. VintageCompressor Ratio & Mix, Standard Compressor DryMix & Hold, Tube Compressor Character & High Ratio, DeEsser side-chain filter & Diff, Magneto II Dual & Oversample, Maximizer Release & Recover, Brickwall Link & Oversample, Noise Gate Hold & Analysis, and more.
- **EQ page (EQ-Eight style)**: band selector (Enc 1), Type / Freq / Q / Gain for the selected band, PreFilter Low-Cut / High-Cut frequency and on/off, PreGain, plus a live magnitude curve (4 EQ bands + PreFilter) with numbered band markers. The curve updates in real time from the Push **and** from changes made in Nuendo.
- **Edit Channel Settings**: now on lower-row button 8 (Channel Strip pages) and Shift+Upper Row (Mix view), in addition to the existing double-press. The button lights white and the on-screen pill turns white while the window is open (bidirectional state feedback).
- **Section colours & bold labels**: each Channel Strip section uses its Nuendo track-strip colour with bold module names, matching the Cubase/Nuendo GUI.
- **Amber Bypass indicators**: bypass pills/LEDs light amber when the section is bypassed (engaged), matching Nuendo's bypass button behaviour.
- **Lower-row LED sync**: lower-row button LEDs follow the on-screen toggle states on the Channel Strip overview and all sub-pages.
- **Piano scale mode**: a new *Piano* option in the Scale selector lays the pads out like a piano keyboard — white keys on the lower row of each octave pair, black keys on the upper row, four octaves stacked bottom-to-top, every C highlighted in purple. Rooted at C (root selection is locked in this mode); Octave Up/Down shifts the whole four-octave block.

Technical

- Strip slot DA exploration (up to 6 slots) with dynamic DA-slot resolution by plugin name — supports custom Cubase strip layouts where module order differs.
- DA param-flip mechanism (CC 73/74 ch8) for binary toggles that Cubase's setTypeToggle handles unreliably (e.g. VintageCompressor Punch, EQ section bypass, EQ band on/off).
- Live host-side parameter feedback via SysEx 0x3D from daInserts.mOnParameterChange, de-duplicated to avoid flooding during mouse drags in Nuendo.
- Edit Channel Settings window state via SysEx 0x3E (separate read binding).
- Dedicated read bindings for PreFilter Phase / section bypass and ChannelEQ bypass so the on-screen state stays in sync.
- SysEx range 0x34–0x3E and ch8 CC 70–78 allocated for strip slot cache, bypass, edit, param-flip, variant probe and live feedback.

Known Limitations

- **Windows — the bridge and Ableton Live cannot use the Push 2 at the same time.** Windows installs the Push 2's WinUSB driver automatically (no Zadig needed); the bridge and Live share that same driver, but only one app can hold the USB connection at a moment — close one to use the other. (macOS is not affected.)
- Channel Strip modules must be **activated manually in Nuendo** first — an empty or disabled strip slot (Gate, Comp, EQ, Tools, Sat, Limiter) is not exposed to the Push 2. Load/enable the module in the Nuendo channel strip before controlling it from the controller.
- Strip slot **variant switching from the Push is not possible** — Cubase locks strip-slot plugin selection; the MIDI Remote API (trySetSlotPlugin / the slot 'Effect Type' tag) is silently rejected. Switch variants in Nuendo's strip-slot menu; the on-screen chip mirrors the active variant in real time.
- Opening a strip-slot plugin's **Edit UI from the Push is not exposed** by the MIDI Remote API (mEdit toggles a flag but Nuendo does not open the window). Use the 'e' button in the mixer console.
- ChannelEQ **bands 2 and 3 are always Peak** filters (Cubase restriction) — the Type encoder has no effect on those bands.
- No audio spectrogram behind the EQ curve — the MIDI Remote API provides no audio stream.
- **Remove the Push 2 'Live Port' and 'User Port' from Nuendo's “In All MIDI Inputs”** (Studio Setup → MIDI Port Setup) or pad notes will be doubled (one copy from the bridge's BridgeNotes port, one direct from the hardware).

Version 1.0.3 — April 2026

New Features

- **Plugin Browser** (Add Device button): browse and load plugins directly from the Push 2. Select an insert slot, scroll through your plugin collection, and load with one press. Uses the DirectAccess Plugin Manager API (Nuendo 15+ / API 1.3).
- **Collection cycling**: switch between Nuendo plugin collections via lower row button 1 in the plugin list view.
- **Plugin Mapper integration**: the Plugin Mapper web server (FastAPI) is now integrated into the bridge and starts automatically. Access it from the menu bar or at http://localhost:8100. Dependencies are optional.
- **DirectAccess insert control**: bypass, edit (open/close UI), and encoder parameter control via DirectAccess for instant response. Fallback to viewer-based control if DA is not available.
- **DA encoder control**: mapped plugin parameters controlled directly via DirectAccess on MIDI channel 9, with display value feedback.
- **Plugin Mapper fuzzy matching**: pedalboard parameter names matched to DA parameter names using word-level fuzzy matching.
- **Record button LED**: 3-state behavior — orange (idle), blinking red (track armed), solid red (recording).
- **Shift+Arrows nudge**: in Volume and Pan modes, Shift+Left/Right shifts the bank by 1 track instead of 8.
- **CC Mode pick-up** (Setup page): new CC Mode page with Absolute and Pick-up options. Pick-up mode prevents parameter jumps by waiting for a direction change before sending.
- **Adaptive text color**: track headers and title bars automatically switch between white and black text based on background luminance. Bold font for headers.
- **Splash screen updated**: removed IAC port reference. Now shows 'Load the MIDI Remote script in Nuendo'.

Technical

- Plugin list transfer via SysEx 0x2C/0x2D, queued in mOnIdle (10 entries/tick).
- Plugin load via CC 80-82 ch8 + trySetSlotPlugin (DA Plugin Manager API 1.3).
- Record state feedback via CC 73.
- Bank shift via CC 38/39 bound to mShiftRight/mShiftLeft.
- Plugin Mapper server in daemon thread with dedicated asyncio event loop.
- MIDI channels 8 (0xB7) and 9 (0xB8) allocated for DirectAccess.

Known Limitations

- No Effect (remove plugin) not supported — API 1.3 has no removeSlotPlugin.
- CC pick-up uses direction-change detection (no Nuendo feedback available).
- Plugin Browser requires Nuendo 15+ (API 1.3).

Version 1.0.2 — April 2026

- Setup Page with Aftertouch Mode and Velocity Curve settings.
- MIDI CC Controller page (Shift+Note): 8 assignable CC faders.
- Mute/Solo/Monitor/Rec on Device and Inserts pages.
- Long press / double press on upper row buttons.
- Clear Monitor / Clear Rec (Shift+Mute/Solo).

Version 1.0.1 — March 2026

- Volume stutter fix, peak clip grace period, Send 1 CC conflict fix, CR CC conflict fix, pan offset correction.

Version 1.0.0 — February 2026

- Initial release: Mixer, Sends, Device, Inserts, Transport, Automation, Control Room, Note Input, Touchstrip, Track Management, VU Meters, macOS Menu Bar App.